

SANSKAR USRETAY

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EDUCATION

Vellore Institute of Technology (VIT)

B.Tech – Computer Science (Artificial Intelligence & Machine Learning)

Aug 2022 – May 2026

CGPA: 8.79 / 10.0

TECHNICAL SKILLS

Programming Languages: Python, SQL

Frameworks & Tools: TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, Streamlit, Flask, Git, Docker, Jupyter Notebook, Google Colab

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, XGBoost, NLTK, OpenCV, TensorFlow, PyTorch, Keras

ML/AI Concepts: Supervised & Unsupervised Learning, Classification, Regression, Clustering, CNNs, RNNs, LSTMs, NLP, TF-IDF, Cosine Similarity, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Data Preprocessing, EDA

PROJECTS

ContentRec – Content-Based Movie & Book Recommendation System | Python, Pandas, Scikit-learn, NLTK, Porter

Stemmer, TF-IDF, Cosine Similarity, Streamlit

2025

- Designed and implemented a content-based filtering system on 10,000+ movie and book records; applied Porter Stemmer (NLTK) for text normalization and TF-IDF vectorization to construct a high-dimensional feature matrix for semantic matching.
- Computed pairwise cosine similarity scores across the entire corpus to generate top-10 ranked recommendations per query; achieved Precision@10 of 0.81 and Mean Average Precision (MAP) of 0.76 in offline evaluation against ground-truth ratings.
- Deployed a fully interactive Streamlit web application with real-time inference latency under 100ms; enhanced recommendations by incorporating collaborative filtering signals, improving catalog coverage by 18% over the pure content-based baseline.

SpamShield – SMS & Email Spam Detection System | Python, Scikit-learn, NLTK, TF-IDF, Naive Bayes, Logistic Regression,

Pandas, Streamlit

2025

- Built an end-to-end NLP text classification pipeline on 10,000+ labeled SMS and email samples; performed tokenization, stop-word removal, stemming, and TF-IDF vectorization; benchmarked Multinomial Naive Bayes against Logistic Regression.
- Achieved 98.2% accuracy and 97.6% F1-score on the spam class using Naive Bayes; applied decision-threshold optimization to minimize false negatives, reducing undetected spam by 31% relative to the default classification threshold.
- Packaged the trained pipeline with joblib for reproducible deployment; served predictions via a Streamlit interface with sub-50ms inference latency on standard CPU hardware, supporting real-time spam filtering use cases.

PriceIQ – House Price Prediction & Regression Analysis | Python, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn,

GridSearchCV

2024

- Developed a full regression pipeline on the Ames Housing dataset (1,460 samples, 79 features); conducted EDA, handled missing values via median/mode imputation, applied ordinal encoding and log-transformation to normalize skewed price targets.
- Trained and ensemble-stacked Linear Regression, Random Forest Regressor, and XGBoost models; performed 5-fold cross-validated GridSearchCV hyperparameter tuning, achieving RMSE of 18,420 and R² of 0.91, placing in the Kaggle top 20%.
- Conducted feature importance analysis to identify top predictors including GrLivArea, OverallQual, and Neighborhood; produced residual plots and evaluation dashboards to validate model generalization and interpretability for stakeholder reporting.

CERTIFICATIONS

Deep Learning Specialization

deeplearning.ai • Coursera

2023

Machine Learning Specialization

Andrew Ng • Stanford University / Coursera

2023

TensorFlow Developer Certificate

Google Brain • TensorFlow

2023

ACHIEVEMENTS & PUBLICATIONS

- Published a peer-reviewed research paper on AI and Machine Learning applications in Agriculture in IEEE — one of the world's leading international journals in computer science and electrical engineering.
- Demonstrated leadership and teamwork through active participation in campus clubs including E-Green Club and Anime Club, contributing to event management, community outreach, and cross-functional collaboration at VIT.
- Strengthened data structures, algorithms, and problem-solving skills by completing 100+ coding challenges on LeetCode, HackerRank, and CodeChef, covering topics directly applicable to machine learning and software engineering roles.